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A Survey of the *Anopheles funestus* (Diptera:Culicidae) Group of Mosquitoes from 10 Sites in Kenya with Special Emphasis on Population Genetic Structure Based on Chromosomal Inversion Karyotypes

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ABSTRACT We sampled *Anopheles funestus* Giles mosquitoes from 10 sites in Kenya to determine (1) the different members of this group present at these sites and (2) the population genetic structure of these species based on chromosomal inversion karyotypes with particular reference to the role of the Eastern arm of the Great Rift Valley and geographical distance as barriers to gene flow. Four members of this species group, namely *An. funestus s.s.*, *An. parensis*, *An. leesoni*, and *An. rivulorum*, were found. *An. funestus* was found almost exclusively inside human dwellings while *An. rivulorum* was found almost exclusively outdoors. *An. parensis* was found both indoors and outdoors in comparable proportions. Earlier attempts to collect this group of mosquitoes from two of the sites yielded no mosquitoes, suggesting that numbers of mosquitoes change during the year, sometime reaching zero or very near zero. Significant levels of differentiation were observed among some *An. funestus s.s.* populations studied. Results suggest that the Rift Valley acts as a barrier to gene flow for *An. funestus*, but that levels of differentiation are not associated with the geographical distance between populations. The results are discussed in the light of the fact that inversions may have some adaptive significance to differences in environmental conditions.

KEY WORDS malaria, *Anopheles funestus*, species composition, chromosomal inversions, variability

MALARIA IS A MAJOR HEALTH concern in sub-Saharan Africa, causing an estimated two million deaths annually (World Health Organization 1998). The most important vectors of this disease are *Anopheles funestus* Giles and two members of the *Anopheles gambiae* complex, namely *An. gambiae s.s.* Giles and *An. arabiensis* Patton. *An. funestus s.s.* is a member of the *An. funestus* group, which comprises at least nine species that are difficult to distinguish based on morphological characteristics (Gillies and De Meillon 1968): *An. funestus*, *An. vaneedeni* Gillies and Coetzee, *An. parensis* Gillies, *An. aruni* Sobti, *An. confusus* Evans and Leeson, *An. rivulorum* Leeson, *An. fuscivenosus* Leeson, *An. leesoni* Evans, and *An. brucei* Service. Of these, *An. funestus* is the most anthropophilic. All the other members are believed to be nonvectors, although *An. rivulorum* has also been found to be a minor vector in Tanzania (Wilkes et al. 1996). Other studies showed that *An. vaneedeni* can be experimentally infected with *P. falciparum* in the laboratory (De Meillon et al. 1977), but no evidence of its role in malaria transmission is available to date.

One of the ways in which malaria may be controlled is through the use of transgenic technology (Gwadz 1994, Crampton et al. 1994) For such an intervention to be successful, a thorough understanding of the population genetic structure of the vector populations is essential. Information on the amount of gene flow between populations indicates how genes for refractoriness would move through the populations and thus is important in determining the scale of introduction of such genes. Additionally, information on levels of gene flow is important in predicting the spread of genes associated with the resistance to insecticides by the malaria vectors.

Recently, two populations of *An. funestus s.s.*, one from coastal Kenya and the other from western Kenya, were analyzed for variability based on paracentric chromosomal inversions (Kamau et al. 2002). Results suggest that the two populations are differentiated. This prompted a follow-up study to investigate the rift valley and geographical distance as possible barriers to gene flow between *An. funestus s.s.* populations. Second, an investigation of the vector potential of *An.*

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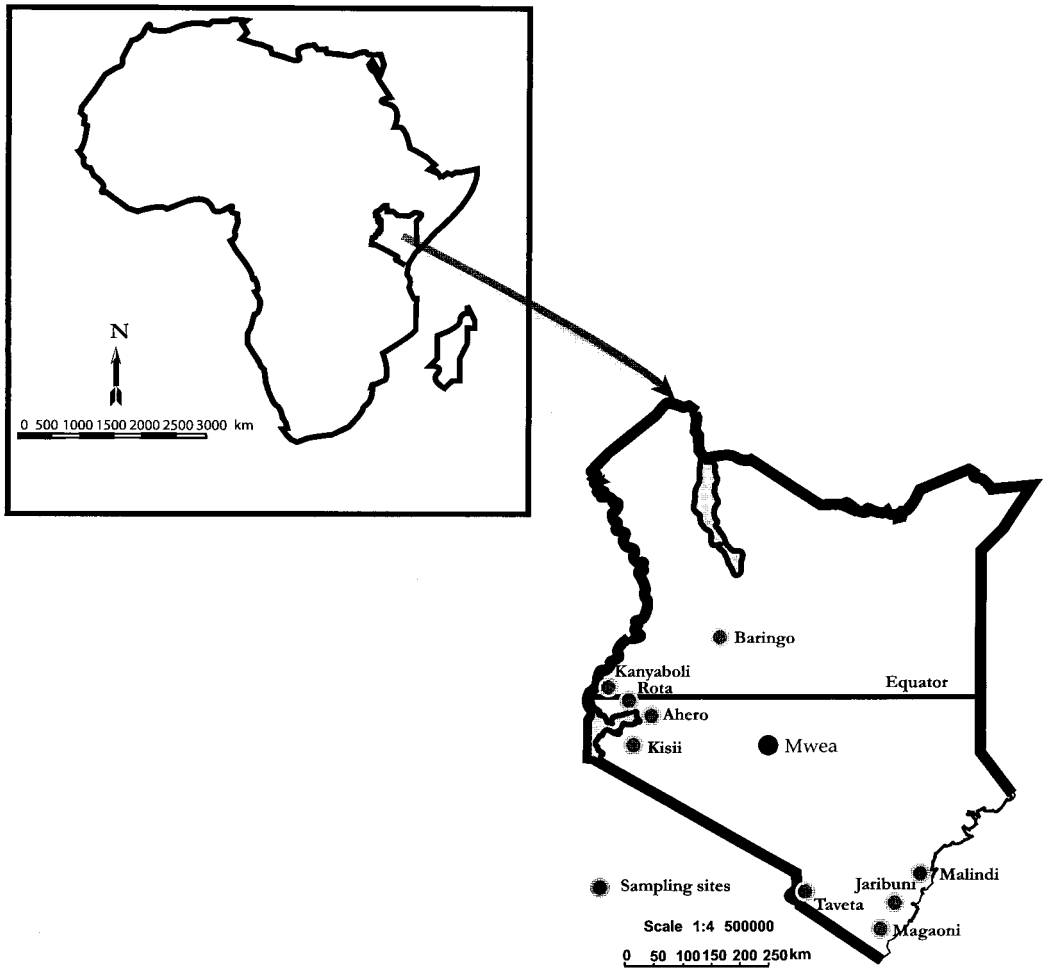


Fig. 1. Map showing the location of the study sites.

funestus s.l. mosquitoes from central Kenya (Kamau et al. 2003) suggest that members of this species group are not important in malaria transmission in this area. *An. parensis* was the main member of the group found, and human blood indices were extremely low, with no mosquitoes found infected with malaria parasites. These results prompted a survey of the different members of the *An. funestus* species group from 10 sites in Kenya with special emphasis on the population genetic structure based on chromosomal inversion karyotypes.

Materials and Methods

Study Sites. Specimens were collected from 10 sites in Kenya: Ahero, Rota, Kanyaboli, Kisii, Baringo, Mwea, Taveta, Magaoni, Jaribuni, and Malindi (Fig. 1). The dates when collections were made are included in Table 1.

Mosquito Collections. Adult female *An. funestus* group mosquitoes were collected from inside houses by aspiration and by the pyrethrum spray collection method as in Githeko et al. (1996). Aspiration was done from house walls in the morning and pyrethrum spray collection in the afternoons. Outdoor collections were also attempted using cow-baited traps and CDC light traps (John W. Hock Company, Gainesville, FL). These were hung out the previous night at around 1800 h and specimens removed before 0700 h. Adult *An. funestus* complex mosquitoes were distinguished from other anophelines based on morphological characteristics (Gillies and De Meillon 1968; Gillies and Coetzee 1987). Half-gravid specimens were immediately preserved individually in Carnoy's fixative in 1.5-ml tubes. Live blood-fed mosquitoes collected during the morning were observed and preserved in Carnoy's once they became half-gravid. All other specimens (unfed, gravid, and blood-fed with ovaries that did not develop) were also preserved in Carnoy's.

Mosquito Species Identification and Chromosome Inversion Genotyping. Half-gravid specimens were removed from Carnoy's fixative, the ovaries were dissected, and squash preparations of the polytene chro-

Table 1. *Anopheles funestus* groups mosquitoes from ten sites in Kenya

Site	Collection date	Indoors/Outdoors	Sample size	Species	% (n)
Ahero	2–5 June, 2000	Indoors	158	*An. funestus*	100% (158)
	"	Outdoors	0	—	—
Rota	6–16 February, 2001	Indoors	62	*An. funestus*	100% (62)
	"	Outdoors	0	—	—
Kanyaboli	3–6 April, 2001	Indoors	70	*An. funestus*	100% (70)
	"	Outdoors	0	—	—
Kisii	9–20 May, 2002	Indoors	81	*An. funestus*	97.53% (79)
				An. parensis	2.47% (2)
		Outdoors	ND		
Baringo	19–20 November, 2000	Indoors	0		
	28 May–1 June, 2001	Indoors	51	*An. parensis*	100% (51)
	"	Outdoors	50	*An. parensis*	100% (50)
Mwea	21–24 November, 2000	Indoors	0	—	—
	12–14 June, 2001	Indoors	414	*An. parensis*	99.76% (413)
				An. lessoni	0.24% (1)
		Outdoors	ND		
Taveta	14–17 November, 2000	Indoors	28	*An. parensis*	96.43% (27)
				An. funestus	3.57% (1)
		Outdoors	ND		
	28 Jan.–1 February, 2002	Indoors	12	*An. parensis*	91.67% (11)
				An. funestus	8.33% (1)
		Outdoors	ND		
	16–19 July, 2002	Indoors	83	*An. funestus*	92.77% (77)
				An. parensis	4.82% (4)
				An. rivulorum	2.41% (2)
		Outdoors	ND		
Magaoni	3–5 May 2000	Indoors	166	*An. funestus*	100% (166)
		Outdoors	ND		
Jaribuni	1–2 May, 2000	Indoors	60	*An. funestus*	100% (60)
	"	Outdoors	50	*An. rivulorum*	98% (49)
				An. funestus	2% (1)
Malindi	21–26 May, 2001	Indoors	2	*An. funestus*	100% (2)
	"	Outdoors	10	*An. rivulorum*	90% (9)
				An. funestus	10% (1)

ND = not done

mosomes were made according to Hunt (1973). The preparations were examined under phase contrast microscopy (model BX50; Olympus, Tokyo, Japan), and paracentric inversion karyotypes were scored using the nomenclature of Green and Hunt (1980) and Green (1982). The fixed inversions on chromosome X were used to distinguish between the different members of the group. *An. parensis* and *An. funestus* are homosequential for Chromosome X and are differentiated using inversion 3c which is only present in *An. parensis.* Specimens that were not suitable for preparation of polytene chromosomes (unfed, blood-fed, gravid) were identified using the recently developed rDNA polymerase chain reaction (PCR) assay for members of the *An. funestus* group of mosquitoes; this assay is based on species-specific primers in the ITS2 region on the rDNA to identify the five most common members of this group namely *An. funestus, An. vaneedeni, An. rivulorum, An. leesoni,* and *An. parensis* (Koekemoer et al. 2002). The different size fragments characteristic of each of these species were distinguished after electrophoresis in ethidium bromide-stained agarose gels. Most of the specimens collected outdoors were identified using this method because they were not at the correct ovarian development stage for identification using chromosomal inversions.

Data Analysis. The distribution of inversion frequencies was tested for conformation to Hardy-Weinberg expectations using Wright's *F* method and the significance of F_{IS} tested using exact probability test available in GENEPOP version 3.2a (Raymond and Rousset 1995), and the Fisher χ^2 method was used to determine the overall significance of the test for particular inversions across all populations. Levels of differentiation were measured using F_{ST} (Weir and Cockerham 1984) in GENEPOP, taking each inversion system as a locus and each alternative chromosomal arrangement to that inversion as an allele. The significance of these values was assessed using the exact test of contingency tables to test for homogeneity of the distribution of inversion across populations. The effective migration rate (*Nm*) was estimated from F_{ST} according to Wright's(1943) formulation: $Nm = 0.25[(1/F_{ST}) - 1]$. The *Nm* indicates the relative strength of gene flow and genetic drift; observed genetic differentiation is expected to be a result of genetic drift if $Nm < 1$, but if $Nm > 1$, then gene flow overcomes the tendency of genetic drift to cause differentiation (Slatkin 1987). This, however, is not a sharp threshold but a useful characterization of results. To investigate if levels of differentiation were related to geographic distances, the regression of $F_{ST}/(1 - F_{ST})$ on geographic distances was determined (Rousset 1997). The effect of the Rift Valley on differentiation was tested by analyzing population on either side of this Rift system only and then populations on opposite sides.

Table 2. Frequencies of the standard form of the polymorphic inversions for *Anopheles funestus* s.s from Ahero, Rota, Kanyaboli and Kisii (in Western Kenya), Taveta, Magaoni and Jaribuni (at the Coast Province). Expected heterozygosities (H_e), Wright's *F* values and significance are also given

Study site	Ahero	Rota	Kanyaboli	Kisii	Magaoni	Jaribuni	Taveta
Inversion 2+a							
Freq. (2n)[a]	0.926 (284)	0.843 (108)	0.955 (134)	0.890 (100)	0.935 (292)	0.974 (114)	0.976 (84)
H_e	0.1374	0.2677	0.0862	0.1978	0.1221	0.0517	0.0470
F	0.2287*	0.5113*	−0.0469	0.4893*	−0.0696	−0.0270	−0.0249
Inversion 3+a							
Freq. (2n)	0.987 (302)	0.803 (122)	0.926 (134)	0.755 (110)	0.759 (332)	0.793 (116)	0.688 (80)
H_e	0.0262	0.3187	0.1391	0.3738	0.3667	0.3310	0.4351
F	−0.0134	−0.0374	−0.0806	−0.1290	−0.0869	−0.2609*	0.0109
Inversion 3+b							
Freq. (2n)	0.896 (298)	0.713 (122)	0.824 (136)	0.713 (108)	0.649 (316)	0.767 (116)	0.675 (80)
H_e	0.1870	0.4125	0.2928	0.4131	0.4572	0.3603	0.4443
F	−0.1161	0.2388	0.0892	0.1403	−0.0693	0.2068	0.3162
Inversion 5+a							
Freq. (2n)	0.908 (316)	0.565 (124)	0.614 (140)	0.566 (106)	0.395 (314)	0.358 (120)	0.298 (84)
H_e	0.1672	0.4957	0.4773	0.4960	0.4794	0.4637	0.4231
F	0.0508	0.2127	0.1559	0.0014	0.0671	0.0214	−0.0054

[a] Freq. denotes the frequencies of each inversion within each population. In brackets are the total number of chromosomal arrangements scored for particular inversions

* *F* value significant, all other values not significant. F is a measure of the Hardy-Weinberg equilibrium. Positive values denote an deficiency of heterokaryotypes while negative values denote an excess.

Results

Species Composition. Collections made by aspiration targeted *An. funestus* group mosquitoes based on morphological identification, and because collections were performed by relatively experience personnel, only the targeted species was collected. Collections using pyrethrum spray collection yielded comparable numbers of *An. gambiae* s.l. mosquitoes. Table 1 summarizes the species composition of *An. funestus* group mosquitoes from the different sites for both indoor and outdoor collections. The first attempts to collect mosquitoes from Baringo and Mwea in November 2000 yielded no mosquitoes belonging to this group. All specimens collected indoors from Ahero, Rota, Kanyaboli, Magaoni, and Jaribuni were *An. funestus s.s* mosquitoes. All specimens collected indoors from Baringo, and all but one of those collected indoors from Mwea were *An. parensis.* The one odd specimen from Mwea was *An. leesoni.* In Kisii, indoors collections yielded 79 *An. funestus* and 2 *An. parensis* specimens. Of the 123 specimens collected indoors from Taveta, all but 8 were *An. funestus.* Six of these eight were *An. parensis* and two were *An. rivulorum.* No *An. funestus* group mosquitoes were collected outdoors from Ahero, Rota, and Kanyaboli. Outdoor collections from Baringo yielded *An. parensis,* whereas those from Jaribuni and Malindi yielded mainly *An. rivulorum.*

Fifteen percent of the specimens that were tested using the rDNA PCR assay failed to give any result.

Chromosomal Inversion Polymorphism. Most of the specimens collected outdoors were at the wrong ovarian developmental stage (mostly unfed) and were therefore not suitable for chromosome preparation. The 27, 28, and 47 *An. parensis* specimens from indoor collections in Taveta, Baringo, and Mwea, respectively, that were scored for the whole chromosome complement were all fixed for inversions 2c, 2g, and 2f. *An. funestus s.s.* was polymorphic for inversions 2a, 3a, 3b, and 5a in all the populations studied. No inversions, however, were found on chromosome X and arm 4.

Population genetic studies were carried out only on *An. funestus s.s.* because inversion polymorphism was observed in this species only. A total of seven populations were included in this analysis; populations from Baringo and Mwea were all 100% *An. parensis* while the Malindi population yielded an insufficient *An. funestus* sample size, i.e., two from indoor collections and one outdoors.

Conformance to Hardy-Weinberg Expectation. Table 2 gives the frequency distributions for the different polymorphic inversions for Ahero, Rota, Kanyaboli, Kisii, Magaoni, Jaribuni, and Taveta populations. Significant deviations from Hardy-Weinberg expectation were observed for inversion 2a for the Ahero, Rota, and Kisii populations. Deviations were caused by a deficiency in heterozygotes. The overall result for this inversion across all populations using Fisher method also showed significant deviations from Hardy-Weinberg expectation.

Tests of genetic differentiation Pairwise F_{st} values for the individual inversions span the wide range of −0.0102 (corresponding to an undefined but infinitely large *Nm* value) to 0.6270 (corresponding to an *Nm* value of 0.15). Table 3 gives mean pairwise F_{st} values averaged across the different inversions. The population from Ahero was significantly different from all other populations, and the Kanyaboli population was significantly different from all but the population from Kisii. The Kisii population, however, was significantly different only from the Ahero and Jaribuni populations (Table 3). F_{st} and *Nm* values for the pairwise comparisons of populations are given in Table 3.

Effect of Geographical Distance and the Rift Valley on Genetic Differentiation. The regression of $F_{ST}/(1 - F_{ST})$ on geographic distances was performed to determine if levels of differentiation were related to

Table 3. Pairwise Fst values for comparisons between the seven *An. funestus s.s* populations studied. Values are Mean Fst values across all inversions

	Ahero	Rota	Kanyaboli	Kisii	Magaoni	Jaribuni
Rota	0.1902*					
Kanyaboli	0.1223*	0.0247*				
Kisii	0.2070*	−0.0073^{ns}	0.0297^{ns}			
Magaoni	0.2752*	0.0235*	0.0714*	0.0180^{ns}		
Jaribuni	0.3340*	0.0361*	0.666*	0.0310*	0.0076^{ns}	
Taveta	0.4239*	0.0569*	0.1290*	0.0459*	0.0042^{ns}	0.0080^{ns}

Significance of values tested by contingency χ^2 test for homogeneity of inversion frequencies; overall significance determined using the binomial probability test; * = significant, ns = not significant.

geographical distances. The regression was not significant (R^2 = 0.0110, df = 19, P = 0.6595; Fig. 2), suggesting that geographical distance alone is not a barrier to gene flow. The effect of the Rift Valley on differentiation was tested by comparing populations on one side of the Rift and then populations on opposite sides. When all populations were considered, the mean F_{ST} value across all inversions was 0.126, corresponding to an *Nm* value of 1.734 and signifying very restricted gene flow. Restricted gene flow was also observed when only populations on the western side of the Rift Valley, namely populations from Ahero, Rota, Kanyaboli, and Kisii, were analyzed (mean F_{ST} across all inversions = 0.111, corresponding to an *Nm* value of 2.000). However, when populations to the eastern side of the Rift Valley, namely those from Magaoni, Jaribuni, and Taveta, were analyzed, mean F_{ST} was 0.006, corresponding to an *Nm* value of 41.4 and signifying a state of panmixia. Taken together, these results suggest that the Rift Valley has a role as a barrier to gene flow.

Discussion

Almost all studies carried out on *An. funestus* in the past have reported results for the group without distinguishing between the different species. This is most likely because methods for identification that have been available, namely morphological and cytotaxonomic identification, are time-consuming and cumbersome. Morphological identification requires rearing of field collected specimens in the insectary because characteristics in immature stages are used (Gillies and Coetzee 1987). In addition, there is considerable overlap in the characteristics used. Cytotaxonomic identification (Hunt 1973, Green and Hunt 1980) requires that specimens be at the half-gravid stage; thus, unfed, fully fed, fully gravid, and male specimens cannot be used. This technique also requires some degree of expertise. Koekemoer et al. (2002) have recently developed a PCR assay that identifies five of the most commonly found members of the *An. funestus* group, making the identification of members of this group easier and with minimal amounts of DNA such as that from two to three dried mosquito legs.

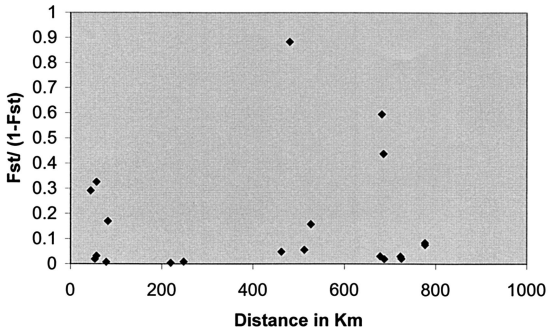


Fig. 2. The regression of Fst(1-Fst) on geographical distance.

Four members of the *An. funestus* group namely *An. funestus s.s.*, *An. parensis*, *An. leesoni*, and *An. rivulorum* were found in our study. *An. funestus s.s.* was found almost exclusively resting indoors in human dwellings while *An. rivulorum* was found in outdoor collections only, except in Taveta where this species accounted for a mere 2.41% of indoor collections. *An. parensis*, however, was found both inside human dwellings and in outdoor collections. *An. funestus s.s.* is the only member of this group that has been confirmed as a major vector of malaria, with *An. rivulorum* having only a minor role (Wilkes et al. 1996). The finding of *An. parensis* resting indoors in large numbers in Baringo, Mwea, and Taveta, and in much lower proportions in Kisii, raises the important question of whether it is a malaria vector as well. Results from a recent study, however, suggest that if *An. parensis* plays any role in the transmission of malaria at all, this role is only a minor one as the human blood index was low and no specimens were found infected with malaria sporozoites (Kamau et al. 2002b). Earlier studies also found no *An. parensis* mosquitoes infected with sporozoites (Gillies and De Meillon 1968, Hargreaves et al. 2000). No *An. funestus s.l.* mosquitoes were collected outdoors from Ahero, Rota, and Kanyaboli, despite finding *An. funestus s.s.* indoors, suggesting that this highly anthropophilic member of the group may be the only one present in this area. This contrasts with Jaribuni, where *An. funestus s.s.* was found indoors and *An. rivulorum* outdoors. Earlier attempts to collected *An. funestus s.l* group mosquitoes from Baringo and Mwea yielded none, suggesting that there are fluctuation in the numbers of these mosquitoes through the year. Although these first attempts to sample *An. funestus s.l.*

mosquitoes in Baringo and Mwea corresponded to the traditional short-rains season, the below-average, and at times failing, patterns of rains in Kenya in recent years does not allow us to categorically describe those sampling time-points as such. Fluctuations in proportions of the different members of the species were indeed evident in Taveta, where a total of three samples were taken: two during two different "short-rains" season and the third at the beginning of the "long-rains" season. While the "short-rains" collections yielded 96.43% and 91.97% *An. parensis*, the "long-rains" collections yielded 92.77% *An. funestus*, suggesting that members of this species complex do not play an important role in malaria transmission during the short rains.

Approximately 15% of specimens did not amplify by the PCR method. This may have been most likely because of degradation of DNA associated with problems in storage or problems in the DNA extraction process. Another possible explanation for this lack of amplification is that the specimens that failed to amplify belong to a species other than the five that can be identified using the PCR assay, namely *An. funestus*, *An. vaneedeni*, *An. parensis*, *An. rivulorum*, and *An. leesoni; An. confusus*, a member of this species group has been recorded from Kenya (Gillies and De Meillon 1968). This is, however, unlikely given that no members of this species were identified using cytogenetics.

Previous studies have suggested that the Rift Valley may act as a barrier to gene exchange between *An. gambiae* and *An. funestus s.s.* populations located on different sides of the Rift (Kamau et al. 1998, 1999, 2002a, Lehmann et al. 1998, 1999, 2000). The theory advanced in invoking the Rift Valley as a barrier to gene flow is that human settlement on the high elevation temperate plateaus is very sparse because they are unsuitable for agriculture; *An. gambiae* and *An. funestus s.s.* are highly domesticated species because of their highly anthropophilic nature, and the sparse human population acts to create discontinuity in the populations of these mosquitoes. Our results support the conclusion that the Rift Valley acts a barrier to gene flow for anthropophilic mosquito species.

Geographical distance has also been suggested as another possible barrier to gene flow in *An. arabiensis* (Donnelly and Townson 2000), *An. darlingi* (Conn et al. 1999), and *An. albimanus* (De Merida et al. 1999) for populations separated by >200 km. The results here suggest that this may not be the case for *An. funestus s.s.*; populations that are more closely located were not more closely related genetically and vice versa. This suggests that the population genetic structure of *An. funestus* in the study did not follow the model of isolation by distance. Geographical distance alone may not be the major barrier to gene flow for *An. funestus s.s.* within distances of between 45 and 780 km.

It is possible that several different factors are responsible for the observed effects. However, the nature of inversions as markers of genetic variability cannot be ignored in this discussion. The use of genetic markers in population genetic analysis is based on the assumption that they are neutral and therefore not under selection pressure. Studies have suggested that this may not be the case for inversions. In *An. gambiae*, for example, inversions have been found to be associated with phenotypes such as sporozoite infection rates (Petrarca and Beier 1992) and resting behavior and adaptation to seasonal changes in humidity (Coluzzi et al. 1979). In *An. funestus* as well, possible associations between inversions and vectorial capacity have been found (Lochouarn et al. 1998, Costantini et al. 1999). We were unable to test for possible associations between sporozoite infection rates and inversions in the current study because of the low sporozoite infection rates in our collection (only 9 of the 516 specimens tested; 1.74%). It seems, however, that the inversions may have some adaptive significance to differences in environmental conditions. The Ahero population, for example, was significantly different from all the other populations. Ahero is the only rice-growing area included in the population genetic analysis. It has long been known that in rice-growing areas, the only member of the *An. gambiae* complex found is *An. arabiensis* (e.g., Githeko et al. 1994, Mukiama and Mwangi, 1989, Ijumba et al. 1990), even in the presence of apparently suitable larval habitats for *An. gambiae s.s.* Although no studies have been done to investigate why this is so, it is possible that factors such as differences in nutrient content of the soil and the pesticides used in rice farming are responsible for the absence of *An. gambiae s.s.* These unique conditions could result in selection for certain inversion karyotypes that are of adaptive significance in *An. funestus s.s.* It is interesting to note that the Rota population was not significantly different from the Magaoni or Jaribuni populations, despite the geographical distance of over 650 km between them and the Rift Valley separation. It is possible that the similar inversion frequencies are maintained by positive selection pressure caused by similarities in environmental conditions such as larval habitats, temperature, and humidity. Rota and Kanyaboli are, however, both located in close proximity to fresh water lakes (2 and 3 km from Lake Victoria and Lake Kanyaboli, respectively) and are separated by a distance of only ≈53 km; however, the frequency of inversions was significantly different. This suggests that environmental conditions may be different in these two sites, contrary to what one would expect because of their proximity to the fresh water lakes. Recent studies on inversion polymorphisms in *An. funestus s.s.* in West Africa, which found significant differentiation between populations located within distances as short as 12 km (Lochouarn et al. 1998, Dia et al. 2000), may also point to the adaptive significance of inversions to different microenvironmental conditions. However, the degrees of inversion polymorphisms were much higher in the West African studies (Costantini et al. 1999, Lochouarn et al. 1998, Dia et al. 2000) compared with the current study. This situation of greater polymorphism of chromosomal inversions in West Africa compared with Kenya has also been observed in the *An. gambiae* complex, where different chromosomal

inversion forms within the complex have been recognized (Toure et al. 1998).

In conclusion, results of this study underscore the importance of proper and complete identification of members of the *An. funestus* species complex in elucidating the role of this species in malaria transmission. Accurate identification will allow access to critical information on biological and behavioral characteristics that are relevant to the control of the vectors. It is apparent, however, that the interpretation of variations in inversion frequencies in allopatric populations is rather complex in the absence of data on environmental heterogeneities because of the possible role of inversions in adaptation to different environmental conditions. Inversions are, however, useful indicators of selection processes.

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